Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory (`https://assumptionsofphysics.org/resources/briefs`) were checked in Lean, where the formal version differs from the brief, and any problems found in the brief.

Setup

- Lean `v4.36.0-rc1`; Mathlib at commit `2cabef97`; physlib (`master`) and TauCeti (commit `eb4dd24`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 The Born rule is equivalent to the von Neumann entropy

Files.

- File: `briefs/BornRuleEntropy/BornRuleEntropy.lean`
- Library: `BornRuleEntropy`
- Imports: `ForPhyslib.QuantumMechanics.VonNeumannEntropyMixture`
- New `ForPhyslib` material:
  `ForPhyslib/QuantumMechanics/VonNeumannEntropyMixture.lean`: a mixture $\sum_i p_i \left|\psi_i\right\rangle \left\langle\psi_i\right|$ of orthonormal pure states, in any dimension, is a density operator (`isDensityOperator_tsum_smul_proj`) with entropy $\sum_i \eta(p_i)$, $\eta(x) = -x \log x$ (`vonNeumannEntropy_tsum_smul_proj`); the entropy of an equal mixture of two pure states is $h((1 + |\left\langle\psi|\phi\right\rangle|)/2)$, h the binary entropy (`vonNeumannEntropy_mix_proj`).

Setting. E is a complex Hilbert space of any dimension. A pure state is $|\psi\rangle\langle\psi|$ (`proj` ψ) for a unit vector ψ, and an ensemble is a density operator (`IsDensityOperator`: positive, trace class, trace one). The entropy S is a function from operators to $[0,\infty]$, and the von Neumann entropy `vonNeumannEntropy` is $\operatorname{tr}\eta(\rho) \in [0,\infty]$ with the natural logarithm, so the brief's "$S = 1$" (one bit) reads $S = \log 2$. The conditional probability is a function p of two operators; $p(|\psi\rangle\langle\psi|, \rho)$ is the probability of finding ψ having prepared ρ. The conditions are:

- `EnsemblesAndStates` p (EQST): $p(|\psi\rangle\langle\psi|, \rho) \in [0,1]$ for density operators ρ; $p(|\psi\rangle\langle\psi|, t\,|\phi_1\rangle\langle\phi_1| + (1-t)\,|\phi_2\rangle\langle\phi_2|) = t\,p(|\psi\rangle\langle\psi|, |\phi_1\rangle\langle\phi_1|) + (1-t)\,p(|\psi\rangle\langle\psi|, |\phi_2\rangle\langle\phi_2|)$ for $t \in [0,1]$; and $p(|\psi\rangle\langle\psi|, |\phi\rangle\langle\phi|) = 1$ iff $|\psi\rangle\langle\psi| = |\phi\rangle\langle\phi|$;
- `BornRule` p (BR-BORN): $p(|\psi\rangle\langle\psi|, |\phi\rangle\langle\phi|) = |\langle\psi|\phi\rangle|^2$;
- `IsVonNeumannEntropy` S (BR-ENT): $S(\rho) =$ `vonNeumannEntropy` ρ for density operators ρ;
- `MutualExclusivity` S p ([B]MUTEX): for every countable family (ψ_i) of unit vectors, $p(|\psi_i\rangle\langle\psi_i|, |\psi_j\rangle\langle\psi_j|) = 0$ for all $i \neq j$ iff $S(\sum_i w_i\,|\psi_i\rangle\langle\psi_i|) = \sum_i \eta(w_i)$ for all weights $w_i \geq 0$ with $\sum_i w_i = 1$ (the sum of operators converging in operator norm).

Results.

Brief	Lean
Proposition 3, eq. (6), $S(\frac{1}{2}\rho_\psi + \frac{1}{2}\rho_\phi) = h((1+\sqrt{\lambda})/2)$, $\lambda = \lvert\langle\psi\vert\phi\rangle\rvert^2$	`vonNeumannEntropy_mix_proj_eq` (from `vonNeumannEntropy_mix_proj`)
Proposition 3, $p(\phi\vert\psi) = \lambda$ iff $S(\frac{1}{2}\rho_\psi + \frac{1}{2}\rho_\phi) = h((1+\sqrt{\lambda})/2)$	`norm_inner_sq_eq_iff`, for $\lambda \in [0,1]$, with p the Born rule
Fig. 2, the entropy is strictly decreasing in the Born probability	`strictAntiOn_entropyOfBorn`
Corollary 7, $\langle\phi\vert\psi\rangle = 0$ iff $S(\frac{1}{2}\rho_\psi + \frac{1}{2}\rho_\phi) = 1$	`inner_eq_zero_iff`, with $\log 2$ for 1
EQST, BR-BORN, BR-ENT, [B]MUTEX	`EnsemblesAndStates`, `BornRule`, `IsVonNeumannEntropy`, `MutualExclusivity`
Theorem 9, BR-BORN and BR-ENT imply [B]MUTEX	`MutualExclusivity.of_bornRule`
Theorem 9, BR-BORN and [B]MUTEX imply BR-ENT	`IsVonNeumannEntropy.of_bornRule`
Theorem 9, BR-ENT and [B]MUTEX imply BR-BORN	`BornRule.of_isVonNeumannEntropy`, over EQST
Theorem 9	`two_conditions_imply_third`
(used for the last direction) orthogonal states are mutually exclusive	`apply_eq_zero_of_inner_eq_zero`
(used for the last direction) the Born rule on the qubit spanned by ψ and $\chi \perp \psi$	`apply_planeVec_eq`: $p(\psi\vert r\psi + s\bar{\zeta}\chi) = r^2$; from `apply_planeVec` and `apply_halfVec`

Differences from the brief.

- E is a Hilbert space of any dimension; separability is not used. [B]MUTEX quantifies over countable index types (in the universe of E). For BR-BORN + [B]MUTEX $\Rightarrow$ BR-ENT the

eigenbasis of ρ is restricted to the eigenvectors with nonzero eigenvalue, a countable family since the eigenvalues are summable.

- Entropies take values in $[0, \infty]$, not in $\mathbb{R}$: the von Neumann entropy and the Shannon entropy of a countable family of weights can be infinite (see "Problems found").
- In EQST, affinity is assumed only for mixtures of two pure states, which is weaker than the brief's countable mixtures. EQST is used only for BR-ENT + [B]MUTEX $\Rightarrow$ BR-BORN, and there only through $p(\psi|\psi) = 1$, $p \geq 0$ on pure states, and affinity for two pure states. The other two directions do not use EQST. Corollary 8 is not formalized.
- The first argument of p is the projection $|\psi\rangle\langle\psi|$, so p depends on the ray of ψ only.
- In BR-ENT + [B]MUTEX $\Rightarrow$ BR-BORN, [B]MUTEX is used only for pairs of orthogonal states, and only in the direction "Shannon entropy $\Rightarrow$ mutually exclusive". In BR-BORN + [B]MUTEX $\Rightarrow$ BR-ENT only the other direction is used.
- The Bloch-ball argument for BR-BORN (Fig. 3) is replaced by an algebraic one in the qubit spanned by ψ and a unit $\chi \perp \psi$. With $f = p(\psi|\cdot)$, $f(\psi) = 1$ and $f(\chi) = 0$, the convex identities $|u\rangle\langle u| + s(r-s)\,|\chi\rangle\langle\chi| = r(r-s)\,|\psi\rangle\langle\psi| + 2rs\,|e_\zeta\rangle\langle e_\zeta|$ (for $r \geq s$, and symmetrically for $r \leq s$), with $u = r\psi + s\bar{\zeta}\chi$ and $e_\zeta = (\psi + \bar{\zeta}\chi)/\sqrt{2}$, give $f(u) = r^2 + 2rs(f(e_\zeta) - \frac{1}{2})$, and $|e_\zeta\rangle\langle e_\zeta| + |e_{-\zeta}\rangle\langle e_{-\zeta}| = |\psi\rangle\langle\psi| + |\chi\rangle\langle\chi|$ gives $f(e_\zeta) + f(e_{-\zeta}) = 1$. If $d = f(e_\zeta) - \frac{1}{2} < 0$, the state with $r/s = -d$ has $f(u) = -d^2/(1+d^2) < 0$, so $f(e_\zeta) = \frac{1}{2}$ and $f(u) = r^2 = |\langle\psi|u\rangle|^2$.

Problems found in the brief. None that affect Theorem 9; some statements need care.

- EQST takes $S : \mathcal{E}(\mathcal{H}) \to \mathbb{R}$, but in infinite dimension the von Neumann entropy is $+\infty$ for some density operators, and $-\sum_i p_i \log p_i$ can be $+\infty$ for countably many weights. With a real-valued S, BR-ENT and [B]MUTEX cannot hold in infinite dimension, and Theorem 9 is true there only vacuously. The formalization uses $[0, \infty]$.
- The units are not fixed: eq. (2) writes log, while Corollary 7 and Fig. 2 use entropy 1 for two orthogonal states, which needs $\log_2$.
- BR-ENT calls $-\operatorname{tr}(\rho \log \rho)$ "the Shannon entropy".
- The proof of BR-ENT + [B]MUTEX $\Rightarrow$ BR-BORN refers to "the proof of 1" and "the same trigonometric arguments in the proof of 1"; there is no item 1, and Proposition 3 and Corollary 7 are meant. It also says "p must take its extreme values on ψ and ϕ", where ψ and $\psi^\perp$ are meant.
- In the same proof, "Given that the range of p is $[0, 1]$, p must take its extreme values on ψ and $\psi^\perp$. Therefore, ψ and $\psi^\perp$ must be opposite points on the ball" does not say why the affine function $p(\psi|\cdot)$ has its gradient along the axis through ψ and $\psi^\perp$; that two orthogonal states are antipodal already holds in the Bloch ball of EQST. The gradient is fixed because $p \geq 0$ near $\psi^\perp$, where $p = 0$; the formalization makes this explicit ("Differences", last item).